

FEHMARN, Germany

WMO: 100550

Lat: 54.5283N

Lon: 11.0606E

Elev: 28

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	24.4	15.5	12.0	24.6	18.6	14.0	25.8	41.0	44.1	36.9	42.7	9.7	120	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	76.4	66.4	73.6	65.6	71.2	64.6	68.3	73.7	66.7	71.6	65.4	69.9	10.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.7	71.1	64.9	92.1	69.4	63.5	87.8	68.2	32.6	73.5	31.3	71.7	30.3	70.0	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.4	30.5	27.1	DB	18.7	83.6	3.6	3.5	16.1	86.1	14.0	88.2	12.0	90.1	9.4	92.7	(4)
			WB	17.3	71.1	3.3	1.5	15.0	72.2	13.0	73.1	11.2	73.9	8.8	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	35.1	35.3	38.6	45.6	53.4	59.3	64.1	64.4	59.0	51.0	43.2	37.0	(6)
(7)		DBStd	11.77	6.23	5.66	5.34	4.74	4.27	3.98	4.05	3.76	4.11	4.97	5.28	6.04	(7)
(8)		HDD50	2051	461	413	353	149	16	0	0	0	0	49	208	402	(8)
(9)		HDD65	5960	926	832	818	583	361	177	67	57	184	435	653	867	(9)
(10)		CDD50	1655	0	0	0	16	121	279	437	447	271	79	5	0	(10)
(11)		CDD65	88	0	0	0	0	0	5	39	39	5	0	0	0	(11)
(12)		CDH74	222	0	0	0	0	8	26	95	80	13	0	0	0	(12)
(13)		CDH80	23	0	0	0	0	1	4	9	8	1	0	0	0	(13)
(14)	Wind	WSAvg	12.8	13.3	14.2	13.5	11.8	11.4	12.3	11.8	11.7	13.6	13.5	12.9	13.5	(14)
(15)	Precipitation	PrecAvg	22.8	1.8	1.4	1.3	1.2	1.9	2.2	2.3	2.5	1.9	2.0	2.1	2.0	(15)
(16)		PrecMax	28.7	3.5	3.0	2.6	2.8	3.6	4.8	5.3	7.5	6.1	4.1	5.0	3.8	(16)
(17)		PrecMin	15.0	0.0	0.0	0.4	0.1	0.5	1.0	0.7	0.6	0.7	1.1	0.4	0.8	(17)
(18)		PrecStd	3.5	0.8	0.7	0.6	0.6	0.9	0.9	1.3	1.7	1.2	0.7	1.2	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	50.3	55.3	64.4	71.7	77.8	80.7	80.0	75.4	64.6	55.8	51.4	(19)
(20)		0.4%	MCWB	47.3	46.8	48.8	56.4	62.3	66.8	68.5	67.5	63.0	59.7	53.1	49.0	(20)
(21)		2%	DB	46.8	46.7	50.3	58.8	65.6	71.9	76.5	76.0	70.4	61.3	53.7	48.1	(21)
(22)		2%	MCWB	44.7	44.6	46.8	52.9	57.8	64.0	66.7	66.7	62.8	57.6	51.5	46.1	(22)
(23)		5%	DB	45.0	44.5	47.7	55.6	62.9	68.3	73.8	73.1	67.5	59.2	51.9	46.2	(23)
(24)		5%	MCWB	43.2	42.7	45.1	50.6	56.6	62.0	66.0	65.8	61.5	56.1	49.6	44.7	(24)
(25)		10%	DB	43.2	42.8	45.4	52.8	60.3	65.6	71.3	70.8	65.4	57.6	50.1	44.7	(25)
(26)		10%	MCWB	41.8	41.5	43.2	49.0	55.2	60.3	64.8	64.9	60.5	54.9	48.0	43.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	47.2	50.5	57.0	62.7	67.9	71.0	70.1	65.7	60.8	53.9	50.0	(27)
(28)		0.4%	MCDWB	49.2	48.9	53.1	63.2	69.5	74.9	77.5	76.1	70.5	63.5	55.2	50.6	(28)
(29)		2%	WB	45.1	44.9	47.5	53.6	59.6	65.0	68.7	68.2	63.8	58.2	51.8	46.6	(29)
(30)		2%	MCDWB	46.5	46.5	49.7	57.3	63.9	70.2	74.1	73.6	68.7	60.5	53.4	47.6	(30)
(31)		5%	WB	43.3	43.0	45.3	51.3	57.7	62.9	67.0	66.8	62.3	56.6	50.0	44.9	(31)
(32)		5%	MCDWB	44.6	44.1	47.5	54.6	61.4	66.9	71.9	71.3	67.0	58.7	51.6	46.0	(32)
(33)		10%	WB	41.9	41.5	43.3	49.5	55.9	61.1	65.5	65.6	60.8	55.2	48.3	43.4	(33)
(34)		10%	MCDWB	43.1	42.7	45.1	52.1	59.3	64.9	70.3	69.6	64.9	57.4	49.8	44.5	(34)
(35)	Mean Daily Temperature Range		MDBR	5.2	5.2	7.0	9.7	10.2	9.7	9.7	9.2	8.0	6.8	5.3	5.3	(35)
(36)		5% DB	MCDBR	6.1	6.5	11.3	15.3	15.1	15.2	15.0	13.2	12.7	8.9	6.1	5.6	(36)
(37)			MCWBR	5.8	5.7	8.5	10.2	9.6	9.2	7.8	7.0	7.3	6.4	5.5	5.4	(37)
(38)			MCDBR	5.8	6.3	10.8	13.6	13.7	13.8	13.1	11.4	11.6	8.1	6.0	5.7	(38)
(39)		5% WB	MCWBR	5.6	5.8	8.4	9.6	9.6	9.1	7.6	6.8	7.2	6.3	5.7	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.341	0.371	0.407	0.398	0.395	0.414	0.413	0.385	0.361	0.340	0.310	(40)	
(41)		taud	2.300	2.317	2.264	2.214	2.339	2.378	2.338	2.351	2.410	2.424	2.361	2.287	(41)	
(42)		Ebn at Noon	195	233	252	259	269	271	262	255	247	225	187	172	(42)	
(43)		Edh at Noon	18	25	33	39	37	36	37	34	29	23	17	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	153	351	765	1312	1675	1776	1657	1333	963	501	202	111	(44)	
(45)		RadStd	23	59	73	159	185	118	189	93	94	65	26	8	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (63 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	-288	N/A	N/A	(47)

Nomenclature: See separate page